

Physics 4220H: Assignment I

Jeremy Favro (0805980)
Trent University, Peterborough, ON, Canada

September 20, 2026

Problem 1. Derive the four typical boundary conditions given below from (the complete) Maxwell's equations. Please include diagrams as needed (drawn by hand is fine):

$$\begin{aligned}\mathbf{E}_1^\parallel - \mathbf{E}_2^\parallel &= 0 \\ \frac{1}{\mu_1} \mathbf{B}_1^\parallel - \frac{1}{\mu_2} \mathbf{B}_2^\parallel &= \mathbf{K}_f \times \hat{\mathbf{n}} \\ \epsilon_1 E_1^\perp - \epsilon_2 E_2^\perp &= \sigma_f \\ B_1^\perp - B_2^\perp &= 0\end{aligned}$$

Solution 1.

Problem 2. Derive the charge continuity equation,

$$\frac{\partial}{\partial t} \rho(\mathbf{r}, t) = -\nabla \cdot \mathbf{J}(\mathbf{r}, t)$$

from (the complete) Maxwell's Equations.

Solution 2. Ampère's law states that

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

Taking the divergence of both sides of this we get that

$$\nabla \cdot (\nabla \times \mathbf{B}) = 0 = \mu_0 \nabla \cdot \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial}{\partial t} \nabla \cdot \mathbf{E}$$

which can be rearranged to

$$\nabla \cdot \mathbf{J} = -\epsilon_0 \frac{\partial}{\partial t} \nabla \cdot \mathbf{E}. \quad (1)$$

Gauss' law then states that

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0}. \quad (2)$$

Plugging eq. (2) into eq. (1) we get the continuity equation,

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t}.$$

Problem 3.

- Show that, in linear media, the Poynting vector can be given by $\mathbf{S} = \mathbf{E} \times \mathbf{H}$, and that the energy density is $u_{EM} = \frac{1}{2} (\mathbf{E} \cdot \mathbf{D} + \mathbf{B} \cdot \mathbf{H})$.
- Describe in words the conceptual difference between the Poynting vector and the total electromagnetic energy density. Can you give an example of when computing one might be more relevant than the other?

Solution 3.

(a) I'll follow basically the same process we did in class except using Maxwell's Equations in linear media,

$$\begin{aligned}\nabla \cdot \mathbf{D} &= \rho_f \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{H} &= \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}.\end{aligned}$$

Now the work done by an electromagnetic field on a charge in infinitesimal timestep dt (over which a length $d\ell$ is traveled) is $dW = \mathbf{F} \cdot d\ell = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot d\ell$. Noting that $\mathbf{v} = \frac{d\ell}{dt}$ we can rewrite this as

$$dW = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v} dt.$$

Noting then that $\mathbf{v} \times \mathbf{B}$ by definition of the cross product gives a vector perpendicular to $\mathbf{v}$ and $\mathbf{B}$, dotting the output with $\mathbf{v}$ will give zero, hence, dividing through by dt ,

$$\frac{dW}{dt} = \mathbf{E} \cdot q\mathbf{v}.$$

Then since $q = \int_V \rho(\mathbf{r}') dr'$,

$$\frac{dW}{dt} = \mathbf{E} \cdot \int_V \rho(\mathbf{r}') \mathbf{v} dr' = \int_V \mathbf{E} \cdot \mathbf{J} dr'$$

provided $\mathbf{E}$ is position independent in V and since $\rho\mathbf{v}$ is a charge moving, i.e. a current. Now applying Ampère's law in media,

$$\begin{aligned}\mathbf{J} &= \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t}, \\ \frac{dW}{dt} &= \int_V \mathbf{E} \cdot (\nabla \times \mathbf{H}) - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} dr' .\end{aligned}$$

Now to fiddle with the first term we use the vector identity $\nabla \cdot (\mathbf{E} \times \mathbf{H}) = (\nabla \times \mathbf{E}) \cdot \mathbf{H} - \mathbf{E} \cdot (\nabla \times \mathbf{H})$ and obtain

$$\frac{dW}{dt} = \int_V (\nabla \times \mathbf{E}) \cdot \mathbf{H} - \nabla \cdot (\mathbf{E} \times \mathbf{H}) - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} dr'.$$

Now applying Faraday's law to the first term,

$$\frac{dW}{dt} = - \int_V \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} + \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} + \nabla \cdot (\mathbf{E} \times \mathbf{H}) dr'.$$

Now using the bit of a priori knowledge from class regarding the chain-rule differentiation trick we can look more closely at the first two terms. Since $\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M}) = \mu\mathbf{H}$ where $\mu = \mu_0(1 + \chi_m)$ and

$$\frac{1}{\mu} \frac{\partial |\mathbf{B}|^2}{\partial t} = \frac{2}{\mu} \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} = 2\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t}$$

we can rewrite the first term as

$$\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} = \frac{1}{2} \frac{\partial \mathbf{B} \cdot \mathbf{H}}{\partial t}$$

and the second term using the same chain rule approach with $\mathbf{D} = \epsilon\mathbf{E} = \epsilon_0(1 + \chi_e)\mathbf{E}$ as

$$\mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} = \frac{1}{2} \frac{\partial \mathbf{E} \cdot \mathbf{D}}{\partial t}.$$

The main expression then becomes

$$\frac{dW}{dt} = - \int_V \frac{1}{2} \frac{\partial}{\partial t} [\mathbf{B} \cdot \mathbf{H} + \mathbf{E} \cdot \mathbf{D}] dr' - \int_V \nabla \cdot (\mathbf{E} \times \mathbf{H}) dr'.$$

We can fiddle with the last term with the divergence theorem (and pull out the time derivative in the first) to obtain

$$\frac{dW}{dt} = - \frac{1}{2} \frac{d}{dt} \int_V [\mathbf{B} \cdot \mathbf{H} + \mathbf{E} \cdot \mathbf{D}] dr' - \oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{a}.$$

I think at this point we can just look at Poynting's theorem in vacuum and compare the two. Performing such a comparison makes it immediately obvious that the term in the first integral is an analogue to u_{EM} , the electromagnetic energy density and the term in the second integral is the Poynting vector.

Situation where one has more use than the other?

- (b) To my understanding, due to the way the dW/dt equation is written, the u_{EM} term represents the total energy stored in the fields in a volume $\mathcal{V}$. The Poynting vector integral term represents the flow of energy through $\mathcal{S}$, the surface which bounds $\mathcal{V}$.

Problem 4. The Dirac Quantization Condition is a famous rule in quantum mechanics that states that if even a single magnetic monopole exists in the universe, electric charge must be quantized. There is a famously cute (if technically improper) way to “show” this, involving quantization of field angular momentum:

Suppose that you had an electric point charge, q_e , separated by distance R from a magnetic point charge (or monopole), q_m . Put the monopole at the origin, and the electric charge at $\mathbf{R}$. The magnetic and electric fields emanating from these charges would be

$$\mathbf{B} = \frac{\mu_0 q_m}{4\pi r^3} \mathbf{r}$$

and

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q_e}{|\mathbf{r} - \mathbf{R}|^3} (\mathbf{r} - \mathbf{R}).$$

- (a) Calculate the field momentum density, $\mathbf{p}_{EM}$, of this arrangement (known as a Thompson Dipole, btw), as a function of r , and comment on the direction of this momentum density.
- (b) From this, find the “field angular momentum” possessed by this arrangement, and comment on
- the meaning (or strangeness) of the functional dependence of this angular momentum on the distance between the two “charges”,
 - the direction of this angular momentum,
 - whether it is intuitive to you that static “charges” can set up fields arranged with nonzero angular momentum.

[Note, the integral involved is notoriously difficult, usually requiring several shortcuts or tricks. If needed, you may invoke the following result:

$$\int \frac{R}{r} \frac{1}{|\mathbf{r} - \mathbf{R}|} \left[\hat{\mathbf{R}} - \hat{\mathbf{r}} (\hat{\mathbf{r}} \cdot \hat{\mathbf{R}}) \right] dV = 4\pi \hat{\mathbf{R}}$$

- (c) A fundamental requirement of quantum mechanics is that angular momentum be quantized in units of $\hbar/2$. Use this fact, together with your result from (b) to argue for Dirac quantization.

Solution 4.